

TrES-3b

R-0383

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230711 · created 2026-07-18T20:49:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460136.8935 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.76 +/- 2.23 [%]
Orbital Inclination (inc)	86.97 +/- 2.85
Airmass coefficient 1 (a1)	0.54 +/- 0.36
Airmass coefficient 2 (a2)	0.46 +/- 0.34
Scatter in the residuals of the lightcurve fit is	58.1 %
Best Comparison Star	#2 - [164, 264]
Optimal Aperture	2.41
Optimal Annulus	7.25
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.194 $\pm$ 0.057 vs 0.16309 (deviation 0.4 σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 1.55 σ)
Mid-time O-C	-4.2 min
Depth SNR	2.5
Residual scatter	58.1 %

Light curve

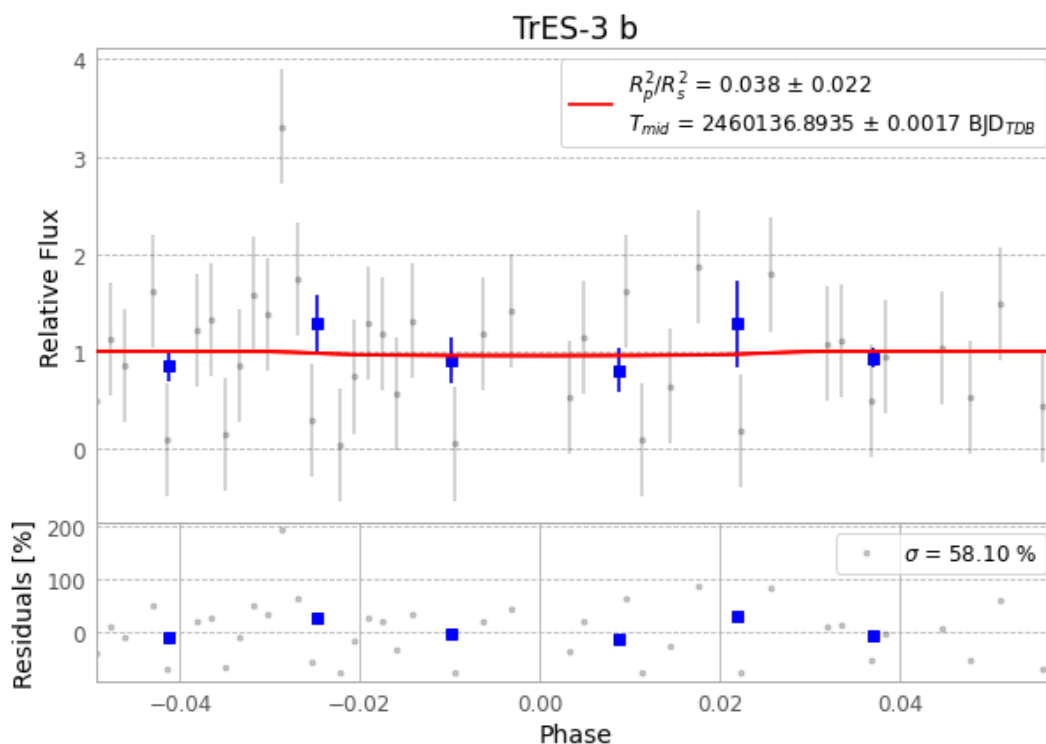


Figure 1 — FinalLightCurve_TrES-3 b_2023-07-11.png (SHA-256 72d885726a32796a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [538, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1471b576348ae666...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

68 FITS frames (45 MB), TRES-3230711074815.FITS → TRES-3230711110915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-TRES-3-230711.log (7fff5667732f...), FinalLightCurve_TrES-3 b_2023-07-11.png (72d885726a32...), FinalLightCurve_TrES-3 b_2023-07-11.csv (aa01e1e54883...)

Compute resources

Wall time 170.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 20:49Z.